



Andrea Zingarofalo

Curriculum Vitae & Studiorum

About Me

Theoretical physicist exploring how fundamental physical laws shape the limits of knowledge, communication, and computation. My current research focuses on device-independent and self-testing methods for certifying quantum resources, a crucial task for establishing trust and security in next-generation quantum technologies. Beyond quantum computing, I am also extremely fascinated by AI and machine learning. I am committed to bridging fundamental science with pressing real-world challenges.

Education

- 2025 – 2029 **PhD in Quantum Information Theory**, *Université libre de Bruxelles*
 - Thesis: *Self-Testing and Device-Independent Verification of Quantum Computation*
 - Advisor: Dr. S. Pironio
- 2020 – 2025 **Diploma di Licenza**, *Advanced School ISUFI* (Excellence Program), 70/70 cum laude
 - ~10% Acceptance rate for the Natural Sciences Class in 2020
 - Modules: Group Theory and Symmetries, Celestial Mechanics, Astroparticle Physics, Applied Physics I–II, Philosophy of Science, ICT for Research, Public Speaking, Scientific Writing
- 2024 – 2025 **Exchange program**, *École normale supérieure*, Master ICFP (International Center for Fundamental Physics) M2, Theoretical Physics
 - Modules: General Relativity, Quantum Field Theory, Statistical Field Theory, Algorithms & Computations
- 2023 – 2025 **MSc in Theoretical Physics**, *University of Salento*, 110/110 cum laude, GPA: 29.8/30
 - Modules: Quantum Field Theory I–II, QCD, Statistical Mechanics, Quantum Computing, Physics of Complex Systems, Data Analysis, Mathematical Methods for AI
 - Thesis: *Exploring the boundaries of quantum correlations: Bell inequalities, convex duality, and self-testing*
 - Advisors: Prof. L. Martina & Prof. J.-D. Bancal
- 2020 – 2023 **BSc in Physics**, *University of Salento*, 110/110 cum laude, GPA: 30/30
 - Electives: Nuclear Physics, Astronomy, Foundations of Physics
 - Thesis: *Study of one-dimensional quantum integrable systems via the Bethe Ansatz*
 - Advisor: Prof. L. Martina
- 2015 – 2020 **High School Diploma**, *Liceo Scientifico V. Lilla*, Oria, Italy, 100/100 cum laude
 - Activities: Student ambassador

Experience

- Oct 2025 – Feb 2026 **Visiting PhD Student**, *Institut de Physique Théorique – CEA Paris-Saclay*
 - Research on verification protocols for quantum computation
 - Supervisor: Prof. J.-D. Bancal
- Nov 2025 – ongoing **AI Model Training Specialist (Physics)**, *Mindrift*
 - Contributed to the training and evaluation of AI models by providing expert-level physics supervision and structured feedback
 - Analyzed model outputs to identify systematic failure modes and guide iterative model refinement

- 2025 – 2029 **Doctoral Researcher**, *Laboratoire d'Information Quantique – Université libre de Bruxelles*
- Research on self-testing and device-independent protocols for certifying components of quantum computation, from multipartite entanglement to error-correcting codes
 - Supervisor: Dr. S. Pironio
- Mar – Jul 2025 **Research Intern**, *Institut de Physique Théorique – CEA Paris-Saclay*
- Derived analytically new Bell inequalities for partially entangled states
 - Implemented SDP relaxations to numerically estimate quantum bounds of Bell inequalities via the Navascués-Pironio-Acín hierarchy
 - Supervisor: Prof. J.-D. Bancal
- Mar – Jul 2024 **Research Intern**, *Mathematical Physics Group – University of Salento (with University of Glasgow)*
- Derived multi-kink solutions of a generalized Fokker-Planck equation via Hirota's bilinear method
 - Developed automated Mathematica scripts for symbolic generation of soliton solutions
 - Supervisor: Prof. L. Martina
- 2019 – Present **Private Tutor** in Mathematics, Physics, and Chemistry for high-school and university students

Publications

Journal Articles

- [1] F. Giglio, G. Landolfi, L. Martina, and **A. Zingarofalo**. Integrability properties and multi-kink solutions of a generalised Fokker-Planck equation. *Journal of Physics A: Mathematical and Theoretical*, 58(16):165202, April 2025.

Science Communication and Outreach

- [2] **A. Zingarofalo**. Quantum correlations: A geometric perspective, 2025. To be published.
- [3] **A. Zingarofalo**. Il conflitto tra meccanica quantistica e senso comune, 2024. Published on Il Chiasmo by Treccani – The Italian Encyclopaedia Institute.
- [4] **A. Zingarofalo**. Il ruolo delle simmetrie nel problema di Keplero. In D. Avaro and F. Decataldo, editors, *Rete di Idee, Udine 2022*, pages 81–100, Udine, 2023. Forum Editrice.

Schools and Workshops

Domain-Specific

- 2025 **Workshop in honor of Ignacio Cirac and Peter Zoller**, Belgian Quantum Initiatives, Brussels, 3 Oct
- 2025 **Interdisciplinary Statistical Physics workshop**, University of Salento, Lecce, 18–19 Sep
- 2025 **Qiskit Global Summer School**, IBM, 7–22 Jul
- 2025 **Summer School “Mathematical Aspects of Quantum Information” (MAQI)**, Université Paris-Saclay and Institut Polytechnique, Paris, 16–20 Jun
- 2024 **Summer School “Theoretical Physics of Fundamental Interactions”**, Scuola Normale Superiore and Scuola Galileiana di Studi Superiori, Pisa, 27–31 May
- 2019 **Physics Summer School**, University of Salento, Lecce, 2–6 Sep
- 2018 **Stage at Asiago Astrophysical Observatory**, INAF (Italian Institute of Astrophysics) and University of Padua, Asiago, 9–13 Jul
- 2016 **Summer School on Astronomy**, INAF and SAI (Italian Astronomical Society), Sant'Agata sui Due Golfi, 5–8 Jul

Others

- 2023 **Summer School “Reframing Sustainability: Bridging Environmental, Energy and Economical Challenges”**, Advanced School ISUFI, Otranto, 10–12 Sep
- 2023 **Winter School “Green and Digital Transition in Europe”**, ENCATC (European Network of Cultural Administration Training Centres), Brussels, 2–7 Dec
- 2019 **University Orientation School**, Scuola Normale Superiore and Accademia Nazionale dei Lincei, Rome, 8–13 Jul

Awards and Scholarships

- 2025 Selected as one of the few Italian members of **Nova** – The Global Top Talent Network that recognizes the **top 3%** of professionals worldwide across STEM, business, and innovation
- 2025 Winner of RIASSU “Rete di Idee”, XIII edition, Catania
- 2025 Selected among the **Top 100 STEM graduates in Italy** (Class of 2025), by Almalaurea, the national consortium of Italian universities
- 2025 Winner of the ISUFI “Research & Innovation” grant for international research mobility
- 2024 Winner of a Cité Internationale Universitaire de Paris scholarship (accomodation)
- 2022 Winner of RIASSU “Rete di Idee”, X edition, Udine
- 2020 Winner of the ISUFI Scholarship at the University of Salento, awarded for academic excellence (full accomodation and a stipend)
- 2018 Senior finalist at XVI Italian Astronomy Olympiads, ranking **among the top 30 out of over 6000 participants** nationwide

Presentations

- 2025 **From Bell’s Theorem to Quantum Technologies via the Device-Independent Paradigm**, RIASSU Rete di Idee, Catania, 23 Nov
- 2025 **Orthogonal Faces in the CHSH Scenario**, *Laboratoire d’Information Quantique – Université libre de Bruxelles*, Brussels, 13 Oct
- 2025 **Orthogonal Faces in the CHSH Scenario** (poster), MAQI School, Paris, 18 Jun
- 2025 **Exploring the Boundaries of Quantum Correlations**, *Institut de Physique Théorique – CEA Paris-Saclay*, Paris, 6 Jun
- 2022 **Il ruolo delle simmetrie nel problema di Keplero**, RIASSU Rete di Idee, Udine, 9 Oct

Technical Skills

Programming and Scientific Computing

- Programming Python (NumPy, SciPy, Matplotlib, pandas), C/C++, Julia, Bash and Javascript; symbolic and numerical computation in MATHEMATICA and MATLAB
- Quantum Libraries Quantum simulation with Qiskit and PennyLane; SDPs and convex optimization using CVXPY, MOSEK, ncpol2sdpa
- Data Analysis ROOT framework for experimental data, and introductory machine learning methods with scikit-learn and TensorFlow
- Workflow L^AT_EX, Markdown, git version control, Inkscape, Microsoft Office; comfortable with UNIX systems, shell scripting, and basic system management

Communication and Teamwork

Collaborated within international research teams; presented theoretical results at seminars and co-authored peer-reviewed work

Languages

Italian (native), English (C1), French (B2), German (A2)

Professional Memberships

IEEE Quantum Technical Community (2025-), Quantum Flagship Community (2025-), Italian Physical Society – SIF (2023-), RIASISSU Excellence Network (2023-2025)

Additional Information

Music Guitar Grade 2 (Distinction), Trinity College of London
Interests Outside academia, I'm a tech enthusiast with a passion for music, history, travelling and geopolitics. Amateur astronomer, speaker at *XXXVII Meeting of Apulian Astrophiles*